

Straight lines - Set 2 - Solutions

Specification MM3.1, M4.9, M4.10.

Answer key

Question	Answer	Topic
1	C	Straight lines
2	B	Straight lines
3	H	Straight lines

Worked solutions

Question 1 LIN-12 • Straight lines • spec MM3.1

The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .

Answer: C

$mx + 4 = 3x - 2 \Rightarrow x = \frac{6}{3-m}$, which is positive iff $m < 3$. Then $y = 3x - 2 > 0$ iff $x > \frac{2}{3}$, i.e. $\frac{6}{3-m} > \frac{2}{3}$, with $3 - m > 0$ this is $18 > 6 - 2m$, i.e. $m > -6$. So $-6 < m < 3$.

Question 2 LIN-13 • Straight lines • spec MM3.1

The distance between the parallel lines $3x + 4y = 5$ and $3x + 4y = 20$ is

Answer: B

The lines are $3x + 4y = c$; the perpendicular direction is $(3, 4)$ of length 5. Distance $= \frac{|20-5|}{5} = 3$. (Alternatively, the point $(0, \frac{5}{4})$ on the first line is 3 from the second.)

Question 3 LIN-10 • Straight lines • spec MM3.1

The line through $(a, 0)$ and $(0, b)$, with $a, b > 0$, passes through $(2, 3)$. Which of the following statements is/are true?

I $\frac{2}{a} + \frac{3}{b} = 1$

II $a > 2$ and $b > 3$

III The smallest possible value of ab is 24

Answer: H

The line is $\frac{x}{a} + \frac{y}{b} = 1$, so I holds. Since $\frac{2}{a} < 1$ and $\frac{3}{b} < 1$, II holds. $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$; setting $a = 2 + t$ gives $\frac{3(2+t)^2}{t} = 3(\frac{4}{t} + 4 + t) \geq 3(4 + 4) = 24$ by AM-GM (equality at $t = 2$, $a = 4$, $b = 6$). III holds.